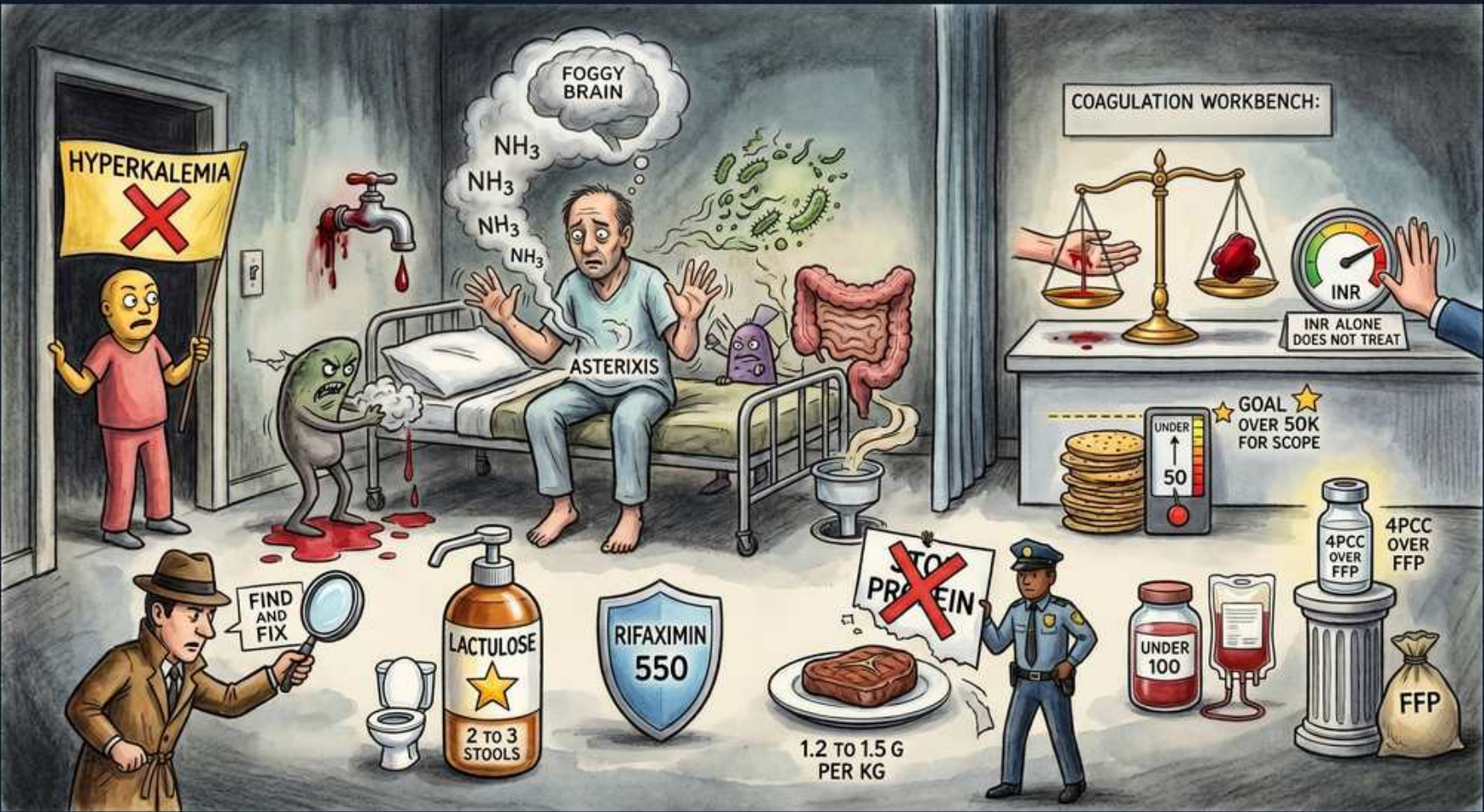


Cirrhosis — Hepatic Encephalopathy & Coagulopathy





1. Hyperkalemia NOT precipitant (X)

WHAT YOU SEE

A figure carrying a yellow flag 'HYPERKALEMIA' with a big red X, far left.

WHAT IT ENCODES

HYPERkalemia is NOT a precipitant of hepatic encephalopathy — it is HYPOkalemia that precipitates HE (low K+ shifts ammonia into cells/brain and increases renal ammoniagenesis). Don't pick hyperkalemia. The real triggers are listed by the detective below.

BOARD TRAP

"Hyperkalemia precipitates hepatic encephalopathy" → wrong; it is HYPOkalemia (plus GI bleed, infection, constipation, dehydration/diuretics, sedatives) that triggers HE.

2. NH3 / ammonia

WHAT YOU SEE

Thought-bubble 'FOGGY BRAIN' with multiple 'NH3' molecules over the patient, center.

WHAT IT ENCODES

Gut-derived ammonia accumulates because the failing liver/portosystemic shunting cannot clear it, crossing into the brain (astrocyte glutamine, swelling). Note: a normal ammonia level does NOT exclude HE and a high level alone doesn't confirm it — HE is a clinical diagnosis. Don't trend ammonia to titrate therapy.

3. Asterixis

WHAT YOU SEE

The patient on the bed with flapping hands labeled 'ASTERIXIS', center.

WHAT IT ENCODES

Asterixis (flapping tremor) is the classic motor sign of HE, a negative myoclonus from impaired posture maintenance. Also seen in CO2 retention and uremia. Graded clinically by West Haven criteria (0-IV).

4. Precipitants: find & fix

WHAT YOU SEE

A detective with magnifying glass and 'FIND AND FIX' sign, lower-left.

WHAT IT ENCODES

Always FIND AND FIX the precipitant: GI bleed, infection (do paracentesis for SBP), constipation, HYPOkalemia, dehydration/over-diuresis, sedatives/opioids, TIPS, dietary protein excess only rarely. Correcting the trigger is as important as lactulose.

5. Lactulose (gold star)

WHAT YOU SEE

A 'LACTULOSE' bottle with gold star and toilet '2 TO 3 STOOLS', lower-center.

WHAT IT ENCODES

First-line HE therapy is lactulose (a non-absorbable disaccharide) titrated to 2-3 soft stools/day (gold star). It acidifies the colon, trapping NH3 as NH4+ for excretion, and is cathartic. Can be given orally, by NG tube, or as retention enema in obtunded patients.

6. Rifaximin 550

WHAT YOU SEE

A blue shield/bottle 'RIFAXIMIN 550', lower-center.

WHAT IT ENCODES

Rifaximin 550 mg twice daily is added to lactulose (especially for secondary prevention/recurrent HE); it is a non-absorbed antibiotic that reduces ammonia-producing gut flora. Combination lactulose + rifaximin lowers recurrence and hospitalization vs lactulose alone.

7. Do NOT restrict protein (X)

WHAT YOU SEE

A steak plate with red X over 'PROTEIN' and '1.2 TO 1.5 G PER KG', center-low.

WHAT IT ENCODES

Do NOT restrict dietary protein in HE — protein restriction worsens catabolism and sarcopenia. Maintain 1.2-1.5 g/kg/day, favoring vegetable/dairy and branched-chain amino acids. Older teaching of protein restriction is obsolete and a tested trap.

BOARD TRAP

"Restrict dietary protein in hepatic encephalopathy" → wrong; maintain 1.2-1.5 g/kg/day protein; restriction causes muscle wasting and is no longer recommended.

8. Rebalanced hemostasis / INR (X)

WHAT YOU SEE

A balance scale 'COAGULATION WORKBENCH' and gauge 'INR ALONE DOES NOT TREAT', top-right.

WHAT IT ENCODES

Cirrhosis is REBALANCED hemostasis: both pro- and anti-coagulant factors fall, so a high INR does NOT reliably predict bleeding. Do NOT transfuse FFP or correct INR based on the number alone in a non-bleeding patient or before low-risk procedures. INR overestimates bleeding risk here.

BOARD TRAP

"Transfuse FFP / correct the INR because it is elevated" → wrong; cirrhosis has rebalanced hemostasis and INR doesn't predict bleeding; don't correct INR alone.

9. Platelet goal >50k

WHAT YOU SEE

A pancake stack with '50' and 'GOAL OVER 50K FOR SCOPE' gauge, right.

WHAT IT ENCODES

For procedures, target platelets >50,000/uL (some endoscopic procedures tolerate lower). Thrombopoietin receptor agonists (avatrombopag, lusutrombopag) can raise platelets pre-procedure and reduce platelet transfusion need. Fibrinogen >120 also matters.

10. 4-factor PCC over FFP

WHAT YOU SEE

A bag '4PCC OVER FFP' next to an 'FFP' bag, lower-right.

WHAT IT ENCODES

When correction IS needed (active bleeding), prefer 4-factor prothrombin complex concentrate (PCC) over FFP — smaller volume avoids volume overload that raises portal pressure and risks rebleeding. Give targeted blood-product support guided by bleeding, not by lab numbers.

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